

PGM Week5 CRF - Team5

OUR BASELINE V1 (Added features: 1~19)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8500	0.8423	0.8461	1668
MISC	0.8086	0.7464	0.7763	702
ORG	0.7826	0.7303	0.7555	1661
PER	0.8627	0.8862	0.8743	1617
micro avg	0.8300	0.8100	0.8199	5648
macro avg	0.8260	0.8013	0.8131	5648
weighted avg	0.8287	0.8100	0.8189	5648

=====

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8199
Delta: +0.0338

⇒ Our Ablation Study is based on this baseline

⇒ Please check the code for detailed feature explanations

OUR BASELINE V2 (Added features: 19 + 6 + 6 + ... – Further Context)

(Context ±3)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8535	0.8453	0.8494	1668
MISC	0.8028	0.7479	0.7743	702
ORG	0.7935	0.7309	0.7609	1661
PER	0.8541	0.8905	0.8719	1617
micro avg	0.8310	0.8125	0.8217	5648
macro avg	0.8260	0.8037	0.8141	5648
weighted avg	0.8297	0.8125	0.8205	5648

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📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8217
Delta: +0.0356

(Context ± 4) $\Rightarrow$ Final Model (# of new Features: 31)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8491	0.8501	0.8496	1668
MISC	0.7960	0.7336	0.7635	702
ORG	0.8079	0.7369	0.7708	1661
PER	0.8526	0.8905	0.8711	1617
micro avg	0.8326	0.8139	0.8232	5648
macro avg	0.8264	0.8028	0.8138	5648
weighted avg	0.8314	0.8139	0.8219	5648

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8232
Delta: +0.0371

(Context ± 5) $\Rightarrow$ No improvement nor degradation

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8576	0.8483	0.8529	1668
MISC	0.7979	0.7422	0.7690	702
ORG	0.7933	0.7417	0.7666	1661
PER	0.8569	0.8850	0.8707	1617
micro avg	0.8322	0.8143	0.8232	5648
macro avg	0.8264	0.8043	0.8148	5648
weighted avg	0.8311	0.8143	0.8222	5648

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8232
Delta: +0.0371

(Context ± 6) $\Rightarrow$ Degraded

... 🚀 Training Custom CRF...

```
=====
                precision    recall  f1-score   support

      LOC      0.8476      0.8471      0.8474      1668
      MISC      0.7960      0.7336      0.7635       702
      ORG      0.7949      0.7327      0.7625      1661
      PER      0.8536      0.8837      0.8684      1617

   micro avg      0.8288      0.8098      0.8192      5648
   macro avg      0.8230      0.7993      0.8105      5648
weighted avg      0.8274      0.8098      0.8180      5648

=====
```

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8192
Delta: +0.0331

Ablation Study (Baseline V1)

- Feature 1 (Word Shape Pattern)

... 🚀 Training Custom CRF...

```
=====
                precision    recall  f1-score   support

      LOC      0.8533      0.8267      0.8398      1668
      MISC      0.7957      0.7379      0.7657       702
      ORG      0.7664      0.7231      0.7441      1661
      PER      0.8542      0.8769      0.8654      1617

   micro avg      0.8220      0.7996      0.8106      5648
   macro avg      0.8174      0.7912      0.8038      5648
weighted avg      0.8209      0.7996      0.8098      5648

=====
```

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8106
Delta: +0.0245

- Feature 2~5 (Extended Affix)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8631	0.8315	0.8470	1668
MISC	0.7927	0.7464	0.7689	702
ORG	0.7641	0.7194	0.7411	1661
PER	0.8393	0.8751	0.8568	1617
micro avg	0.8193	0.8005	0.8098	5648
macro avg	0.8148	0.7931	0.8034	5648
weighted avg	0.8184	0.8005	0.8090	5648

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8098
Delta: +0.0237

- Feature 6~9 (Punctuation & Alphanumeric Flags)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8503	0.8447	0.8475	1668
MISC	0.8031	0.7379	0.7691	702
ORG	0.7780	0.7321	0.7543	1661
PER	0.8588	0.8837	0.8711	1617
micro avg	0.8269	0.8095	0.8181	5648
macro avg	0.8225	0.7996	0.8105	5648
weighted avg	0.8256	0.8095	0.8171	5648

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8181
Delta: +0.0320

- Feature 10, 11 (POS Bigrams)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8532	0.8465	0.8498	1668
MISC	0.8176	0.7536	0.7843	702
ORG	0.7643	0.7243	0.7437	1661
PER	0.8617	0.8825	0.8720	1617
micro avg	0.8263	0.8093	0.8177	5648
macro avg	0.8242	0.8017	0.8125	5648
weighted avg	0.8251	0.8093	0.8168	5648

🇮🇹 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8177
Delta: +0.0316

- Feature 12~14 (Extended Context Before (-2))

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8529	0.8375	0.8451	1668
MISC	0.7885	0.7436	0.7654	702
ORG	0.7667	0.7261	0.7458	1661
PER	0.8565	0.8751	0.8657	1617
micro avg	0.8217	0.8038	0.8127	5648
macro avg	0.8162	0.7956	0.8055	5648
weighted avg	0.8206	0.8038	0.8119	5648

🇮🇹 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8127
Delta: +0.0266

- Feature 15~17 (Extended Context After (+2))

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... 🚀 Training Custom CRF...
=====
                precision    recall  f1-score   support

      LOC      0.8542      0.8393      0.8467      1668
      MISC      0.7876      0.7393      0.7627       702
      ORG      0.7807      0.7351      0.7572      1661
      PER      0.8527      0.8769      0.8646      1617

   micro avg      0.8250      0.8070      0.8159      5648
   macro avg      0.8188      0.7977      0.8078      5648
weighted avg      0.8239      0.8070      0.8151      5648

=====

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1:   0.8159
Delta:       +0.0298

```

- Feature 12~17 (Extended Context)

```

... 🚀 Training Custom CRF...
=====
                precision    recall  f1-score   support

      LOC      0.8520      0.8315      0.8416      1668
      MISC      0.7819      0.7507      0.7660       702
      ORG      0.7653      0.7243      0.7442      1661
      PER      0.8514      0.8609      0.8561      1617

   micro avg      0.8185      0.7983      0.8083      5648
   macro avg      0.8126      0.7918      0.8020      5648
weighted avg      0.8176      0.7983      0.8077      5648

=====

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1:   0.8083
Delta:       +0.0222

```

- Feature 18, 19 (Number + Title, Headline)

```

... 🚀 Training Custom CRF...
=====
                precision    recall  f1-score   support

      LOC      0.8482      0.8375      0.8428      1668
      MISC      0.8067      0.7493      0.7770       702
      ORG      0.7715      0.7339      0.7522      1661
      PER      0.8620      0.8844      0.8730      1617

   micro avg      0.8256      0.8095      0.8175      5648
   macro avg      0.8221      0.8013      0.8113      5648
weighted avg      0.8244      0.8095      0.8166      5648

=====

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1:   0.8175
Delta:       +0.0314

```

- Feature 18 (Number + Title)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8532	0.8435	0.8484	1668
MISC	0.8021	0.7507	0.7756	702
ORG	0.7787	0.7309	0.7540	1661
PER	0.8599	0.8881	0.8737	1617
micro avg	0.8282	0.8116	0.8198	5648
macro avg	0.8235	0.8033	0.8129	5648
weighted avg	0.8269	0.8116	0.8188	5648

📊 Performance Comparison:
 Baseline F1: 0.7861
 Custom F1: 0.8198
 Delta: +0.0337

only +0.0001 improvement (compared to our baseline v1)

- Feature 19 (Headline)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8441	0.8405	0.8423	1668
MISC	0.8009	0.7450	0.7720	702
ORG	0.7734	0.7273	0.7496	1661
PER	0.8603	0.8794	0.8697	1617
micro avg	0.8238	0.8065	0.8151	5648
macro avg	0.8197	0.7981	0.8084	5648
weighted avg	0.8225	0.8065	0.8141	5648

📊 Performance Comparison:
 Baseline F1: 0.7861
 Custom F1: 0.8151
 Delta: +0.0290

- Feature 6~9, 19 (Complex single word features)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8446	0.8405	0.8425	1668
MISC	0.8034	0.7450	0.7731	702
ORG	0.7730	0.7339	0.7529	1661
PER	0.8571	0.8825	0.8696	1617
micro avg	0.8232	0.8093	0.8162	5648
macro avg	0.8195	0.8005	0.8095	5648
weighted avg	0.8220	0.8093	0.8153	5648

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8162
Delta: +0.0301

- Feature 2~5, 12~17 (Context extension features)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8554	0.8333	0.8442	1668
MISC	0.8009	0.7507	0.7750	702
ORG	0.7668	0.7164	0.7407	1661
PER	0.8293	0.8565	0.8427	1617
micro avg	0.8160	0.7953	0.8055	5648
macro avg	0.8131	0.7893	0.8007	5648
weighted avg	0.8151	0.7953	0.8048	5648

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8055
Delta: +0.0194

- Feature 10, 11, 18 (Interaction features)

... 🚀 Training Custom CRF...

	precision	recall	f1-score	support
LOC	0.8542	0.8429	0.8485	1668
MISC	0.8137	0.7593	0.7856	702
ORG	0.7598	0.7200	0.7394	1661
PER	0.8596	0.8788	0.8691	1617
micro avg	0.8242	0.8067	0.8153	5648
macro avg	0.8219	0.8003	0.8107	5648
weighted avg	0.8230	0.8067	0.8145	5648

📊 Performance Comparison:
Baseline F1: 0.7861
Custom F1: 0.8153
Delta: +0.0292

⇒ The reason why the ablation study is based on v1 is because v2 was developed after the ablation study.
We might do the another ablation study for v2... But I don't think it is that necessary.